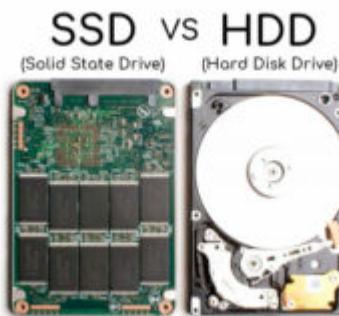


Databases

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I. CHAPTER 1 DATA THE NEW CORPORATE RESOURCE

The importance of storing data can be traced back to ancient times and it is continued to be utilized today by information scientists. Infrastructure continues to grow larger and larger for increasing amounts of data processing.

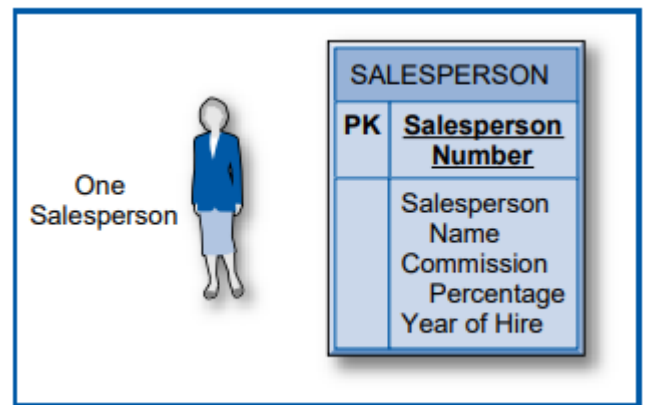


The importance of data led to much development of electromechanical calculating devices and then to modern devices with updated storage options. Businesses gain the opportunity of using large storage data devices yet having large amounts of data became a challenge.

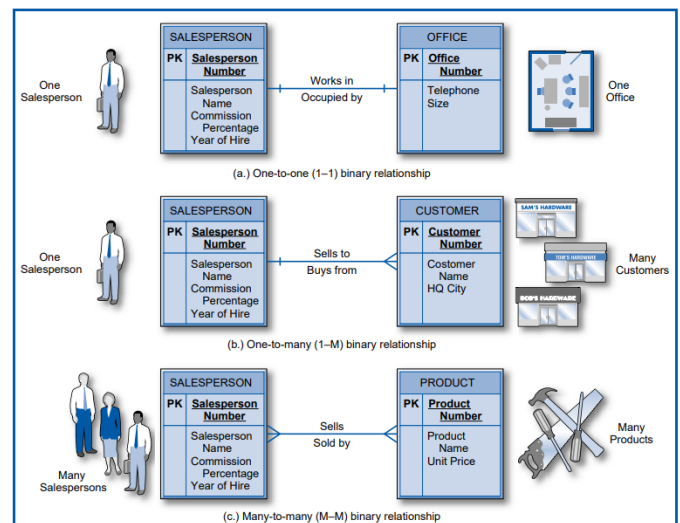
The retrieval of data and backup of data became of great importance. Businesses would need to access data simultaneously instead of individual cases led to increase of data management. Teams of people would require accurate and maintained data with security for accessing data daily. Data is becoming a critical resource for business and individual use. Managing data and refining it is a complex task that led to the developing new ways for refining data with software known as database management.

II. CHAPTER 2 DATA MODELING

Data modeling consist of a diagramming technique that can contrasts entity-relationship type models. The entity-relationships are made of diagram data types that hold attributes to describe an entity or scenario within various environments. The entity is an objective environment that allows the user to create an object which can describe certain attributes pertaining to an activity that object is used for.



An entity-relationship begins with a diagram that creates objects with attributes and unique identifiers. The attribute is important when an object is created within an environment since it will describe what the object activity is. The unique identifiers are great for implementing consistent balance between other object relationships in a diagram.



A relation in an entity can organize multiple associated objects together with in a diagram. A popular example is when a Parent is associated with their children, or a tree is associated

with an apple. The relationships between two object types inside an environment is important to create accurate diagrams.

Binary relationship entities insist that two entities are related, but unary relationships insist that two entities are the same. In a ternary relationship, three entities are in relation to the main object .

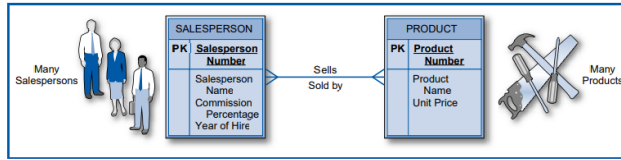


FIGURE 2.2
A binary relationship

III. CHAPTER 3 THE DATABASE MANAGEMENT SYSTEM CONCEPT

Data integration references user control data manipulation, the data is pieced together with other related data within an information system. Data files are popularly used with data integration to organize data within one container including multiple attributes. The attributes can include descriptive subjects pertaining to the specific data file. An example would be a word file that contains information describing a person's identification or physical location similar to identifications.

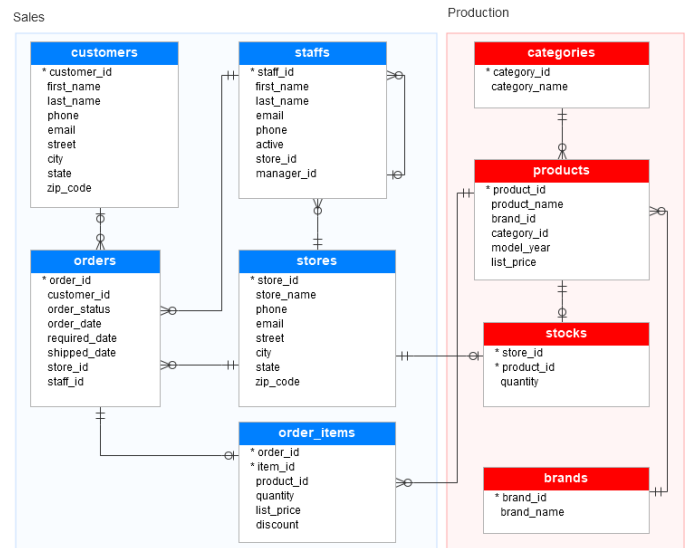


Data redundancy would be similar to data integration yet, the data is duplicated or stored more than once with the same information. Having duplicate information in one file container can be redundant. Two files with the same information can be redundant as well. It is in best practice to keep files organized with separate information in one file when combining data information. Database management would be difficult to manage with redundant files having the same information.

IV. CHAPTER 4 THE DATABASE MANAGEMENT SYSTEM CONCEPT

Data management revolves around two aspects: data definition and data manipulation. Data definition is an operationalized instructed software on what tables will be in the database, which attributes are applied to the database, and indexing of the information. Data manipulation requires special

language codes for communication between user commands to the database management system. A language that can be used with database management communications is a Structured Query Language (SQL) since it both uses data definitions and data manipulation practices.



Data retrieval is done using SQL SELECT commands for creating fundamental databases and ideas. SQL commands could be considered declarative instead of procedural. The data can be specified instead of using a logical sequence of steps that guide a database system in finding data.

SQL became the standard for most database management systems for creating relational information. SELECT commands can use direct query mode or customized language programs for embedded mode. The rows and tables can be modified using SELECT commands for combining other functions inside the database.

Performance is an important aspect of data retrieval when dealing with a higher level of database storage and management. A relation query would require software-based functionalities to solve each relation query. Query optimizers work towards incorporating database techniques.

V. CHAPTER 5 THE RELATIONAL DATABASE MODEL: INTRODUCTION

The relational database is considered to be a data structure that formats linear data files based on relations and attributes. The relational database could include tables holding the attributes within organized by rows. The rows would hold the record information about the object sorted into field columns. The fields can be arranged in a specific order based on the relation of data. The relational database would then be a collection of relations once sorted. The databases would use a relation key such as primary keys to signify the main relation.

Attribute (or Column or Field)				Relation (or Table or File)
Student Number	Student Name	Class	Major	
03657	Robert Shaw	Senior	Biology	Tuple (or Row or Record)
05114	Gloria Stuart	Freshman	English	
05950	Fred Simpson	Junior	Mathematics	
12746	W. Shin	Junior	English	
15887	Pedro Marcos	Senior	History	
19462	H. Yamato	Sophomore	French	
21682	Mary Jones	Freshman	Chemistry	
24276	Steven Baker	Sophomore	History	

The primary key is an attribute to describe the object with unique values about that very object. Candidate keys are used if a relation has more than one attribute. Candidate keys are made before primary keys since the object is subject to more than one attribute, however, one candidate key must be chosen as the primary key afterward. Once the candidate key has been chosen to be primary then the rest of the candidates become alternate keys. This process is very important in business since the primary key is chosen since it will be best suitable for applications that use that very primary key in databases. Some scenarios have a collection of relations that make a relational database with a primary key serving one relation holding groups of attributes and then could appear in another relation. If a primary key appears in another relation then it is called a foreign key since it is a duplicate key but in another environment within one database. A primary key in one relation area could be considered different in another related area yet treated as foreign since the attributes wouldn't be as primary in another relation.

VI. CHAPTER 6 THE RELATIONAL DATABASE MODEL: ADDITIONAL CONCEPTS

The concept of a unary-to-many relationships involves an entity pointing to one relation in the database. Some of the entities hold additional attributes related to the main object created in the database. One example would be a salesperson reports to exactly one sales manager although each salesperson who serves as a sales manager typically has several salespersons reporting to him (Wiley, Sons, 2012). In this situation it is many sales personal reporting to one manager. The entity would be the salesperson and the relational object would be the sales manager.

A one-to-many relationship can be constructed with only one entity type since the entity will be under one relational object. It is good to keep in mind that one-to-many constructed databases may have entity occurrences that involve many levels in a hierarchical arrangement (Wiley, Sons, 2012). Some databases may have to include the relational object in relation to another object that holds its own entities.

(a) SALESPERSON relation				
Salesperson Number	Salesperson Name	Commission Percentage	Year of Hire	Office Number
137	Baker	10	1995	1284
186	Adams	15	2001	1253
204	Dickens	10	1998	1209
361	Carlyle	20	2001	1227

(b) CUSTOMER relation			
Customer Number	Customer Name	Salesperson Number	HQ City
0121	Main St. Hardware	137	New York
0839	Jane's Stores	186	Chicago
0933	ABC Home Stores	137	Los Angeles
1047	Acme Hardware Store	137	Los Angeles
1525	Fred's Tool Stores	361	Atlanta
1700	XYZ Stores	361	Washington
1826	City Hardware	137	New York
2198	Western Hardware	204	New York
2267	Central Stores	186	New York

(c) CUSTOMER EMPLOYEE relation			
Customer Number	Employee Number	Employee Name	Title
0121	27498	Smith	Co-Owner
0121	30441	Garcia	Co-Owner
0933	25270	Chen	VP Sales
0933	30441	Levy	Sales Manager
0933	48285	Morton	President
1525	33779	Baker	Sales Manager
2198	27470	Smith	President
2198	30441	Jones	VP Sales
2198	33779	Garcia	VP Personnel
2198	35268	Kaplan	Senior Accountant

(Continues)

A many-to-many relationship brings the entity occurrences hierarchical concept which involves a single entity occurrence related to other occurrences. The relational concept of many-to-many occurrences involves multiple parts ultimately creating one larger occurrence together. The key concept in this scenario database would be smaller entities coming together to form a larger entity all-in-all.

In a relational database many-to-many occurrences could be constructed with many entities having their own sectioned rows in relation with other entities to make the overall object set. The many-to-many relationship could require creating additional relationships with multiple attributes. The domain of each column may consist of an attribute-based value related to the object. The objects are assembled to handle comprised entities.

VII. CHAPTER 7 LOGICAL DATABASE DESIGN

Logical database design is the process of decided how to arrange the attributes of the entities in a given business environment into database structures, such as the tables of relational database.

Physical database design is the process of modifying the logical database to improve performance.

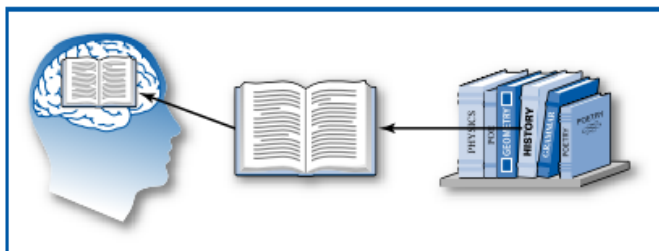
```
CREATE TABLE SALESPERSON
(SPNUM CHAR(3) PRIMARY KEY,
SPNAME CHAR(12)
COMMPERCT DECIMAL(3,0)
YEARHIRE CHAR(4)
OFFNUM CHAR(3));
```

The main logical database techniques used were data normalization to formalize database design. Today, Entity-Relationship model and other diagramming tools are used with combining data normalization techniques to convert database structures and to check full-scale databases from E-R diagrams. The process includes organizing attributes into tables to prevent redundancy.

- One-to-one binary relationship. - Two entities are combined into one relational table.
- One-to-many binary relationship. - Unique identifier of the entity is placed as a foreign key. in the table representing the entity.
- Many-to-many binary relationship. - This conversion requires multiple tables where each entity holds its own attributes but with no foreign keys.
- One-to-one unary relationship. - One entity type involved with the same role group requires one table.
- One-to-many unary relationship. - Similar to unary definition and process.
- Many-to-many unary relationship. - Requires two tables without any foreign keys. A second table is created equal to the number of entities.
- Ternary relationship. - A primary key is created in combination with unique identifiers of multiple entities and attributes.

VIII. CHAPTER 8 PHYSICAL DATABASE DESIGN

Data operates all around us and holds important information about every person in the world. Data keeps track of information for many people for reference and is used to apply objects within a system. The objects and events can be categorized with attributes that are applied to a structure of records.

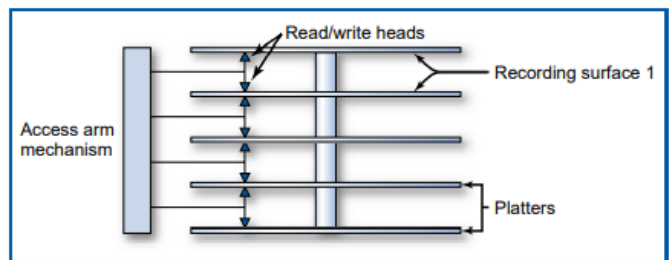


Storing data in computers consists of four basic operations for optimal performance: retrieving, inserting, deleting, and

updating data. Most database storage applications require data operations to ensure daily access to stored data.

Disk devices are the most popular used storage options today since speed and performance are offered in most data disks. Disk users are given access to sequential and direct operations for the data. Magnetic engineering helps store data on the disks and is constructed of a multi-layer system. Data is stored on each layer for the circulation of data in the disk.

The arrangement of data on disks is constructed with a file system that allows data to be retrieved using an access method. A disk access method includes two popular options: indexing and hashing. Linear indexing holds two columns: an ordered list of the identifiers of the records being indexed, each of which is associated in the second column with its physical location on the disk. Practical arrangements could include B+-tree methods, in which the index is constructed in a hierarchical arrangement. Hashing helps arrange records on the disk based on calculations of each record identifier. The retrieval of data would be based on each record identifier.



Physical database design techniques use logical designs and logical design changes. Logical designs change by adding external features such as indexes, reorganized data on the disk, and splitting a table into multiple tables. Attributes are applied tables and the same attributes can be changed by combining new tables or adding new tables.

IX. CHAPTER 9 OBJECT-ORIENTED DATABASE MANAGEMENT

Object-oriented data modeling terminology involve entities described with objects and events correlating to a given environment. Object holds the information about an advanced data structure with entities attributes while containing program coding for operations using entities attribute values. The object entities are organized with the object class since the records and file information are bounded to the class.

The advanced relationship generalization or specialization is a relationship that recognizes different entities that can be subdivided into smaller groups specialized with those given entities. The benefit of generalization entities are the ability to hold common characteristics for those smaller created groups to better organize the database. A group can be located since each group may have unique characteristics.

Attributes can be inherited in a generalization hierarchy by indicating entities common attributes using class. The class coordination can allow an object class to inherit all the attributes from another object within a general hierarchy.

Operations hold different actions: constructor, query, and update that help create additions to inheritance and polymorphism. The constructor operation creates a new

instance of a class. The query operation returns data about the values of objects attributes but keeps those objects the same. An update operation helps renew objects attribute values within the class. Polymorphism can modify or refine the three operations as they are inherited downwards in the database. The operations all work together to organize and manage a database.

X. CHAPTER 10

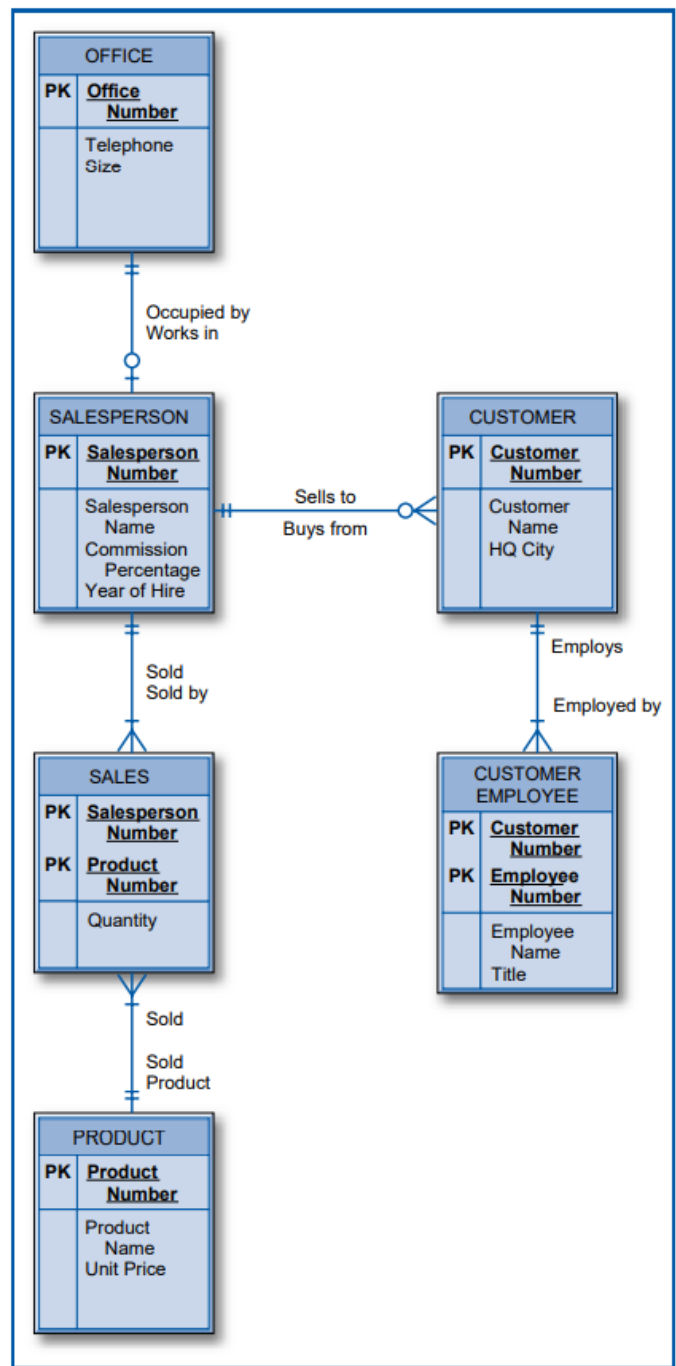
Data Administration involves the process of planning data policies and analyzation. The analysis function is responsible for setting data policies and standards for proper business data resources. The data can be related to different categories like accounting or providing liaison support to system analysts during application development.

Database administration involves functions that result in operational structure and responsible for daily monitoring of the business system. The monitoring could include active databases or support to program designers during development.

Data dictionaries hold a full database pertaining to data. The dictionary structure could result to data repositories when holding certain data categories made by the administrator. Usually, metadata is used to make up data dictionaries for the whole database structure.

The responsibilities of data administration can be assigned based on the role as a data analyst. The corporate environment could include data coordination as a requisite to handling data. Data coordination is essential for employees looking for the same or similar information in a business setting. Data planning also determines which data will be needed to support the team. Data standards make a certain normalcy that plays into interactions between supply practices.

Metadata helps towards the function of how data is handled. Metadata is the same as data dictionary or data repository since they all handle data about data. Passive data dictionary is one of the methods to document data. It is a self-contained database for the Information System environment. An active data dictionary is based on real-time data interaction that holds similar functions to passive databases. Databases are capable of being paired as dictionary relationships to help aid new developments or projects in business. The current database practice would be data repositories that handle larger databases in Information systems.



XI. CHAPTER 11 DATABASE CONTROL ISSUES: SECURITY, BACKUP AND RECOVERY, CONCURRENCY

Data security is protecting the data from theft, from malicious destruction, from unauthorized updating. Good data security is absolutely critical to every company and organization. Data security can affect a company's ability to continue day to day functions.

1. Types of data security breaches:

- unauthorized data access - someone who obtains data that she is not authorized to see
- unauthorized data or program modification - someone changes the value of stored data that they are not entitled to change
- malicious mischief - someone can corrupt or even erase some of a company's data; hardware can be damaged, making data unusable
- unauthorized computer access - hacking or gaming access from outside company
- intercepting data communications - could be wiretapping, data can be stolen while it is being transmitted
- stealing disks or computers - could involve flash disks and CDs all have the potential of being stolen, laptop computers being stolen, even desktops as well being stolen
- computer viruses - a malicious piece of software that is capable of copying itself and spreading from computer to computer
- damaging computer hardware - it could be deliberate or accidental such as fires, coffee spills, hurricanes, or disgruntled employees
- the importance of backup and recovery - if something goes wrong with the computer data it is always good to have a backup to correct the data after corruption
- backup copies and journals - backing the database and maintaining the computer journal is important which involves scheduled backups and tracking any changes to the computer
- forward recovery - recreation of the backup table and creating a log for applying changes to the backup table since the last backup
- backward recovery - allow recovery program to proceed backward through the log by restoring each updated data value in database to its before image until it reaches the point where the error was made
- duplicate or mirrored databases - two copies of the entire databases are maintained, and both are updated simultaneously
- disaster recovery - rebuilding an entire information system or significant parts of one, after a catastrophic natural disaster such as: hurricanes, tornadoes, earthquakes, building collapse, or a major fire

Types of data security measures:

- Physical security of company premises - recommended not to put computer in the basement because of risk to floods or other damaging scenarios
- Controlled access to the computer system - could involve setting a defense plan with identification tag or password
- Controlled access to the database - restrict access to specific data so that only specific people can retrieve or modify it, some systems have such controls in the operating system or in other utility software, an additional layer of passwords may also be introduced
- Data Encryption - Data is changed, bit-by-bit or character-by-character, into a form that looks totally garbled. Data can be stored, transmitted, or encrypted. To be used, data must be decrypted.
- Anti-virus software - software used to combat computer viruses by locating virus signatures or monitoring the computer itself for malicious activity
- Firewalls - software or a combination of hardware that protects a company's computer and its data against external attack via data communications lines

Backup and Recovery:

XII. CHAPTER 12 CLIENT/SERVER DATABASE AND DISTRIBUTED DATABASE

A client/server database is for personal computers connected on a local area network to help locate data.

The client/server database involves a local-area network which is an arrangement of personal computers connected by communications lines. Additional components needed for a local-area network can be utilized or shared by the personal computers can be other, often more powerful server peripheral and computer device type. Lastly, a gateway computer on the local area can link the network and its personal computers to other local area networks, to one or more mainframe computer or the internet.

One of the main advantages of a local-area network is the ability to share resources, then certainly one type of resources to share is data contained in databases.

Software was made to ensure information is secure within the server and the distribution of information can be transparent. The important of data transparency is securing information from specific users.

The two-tiered client /server arrangement involves the file server approach when a client computer on a local-area network needs to query, update, or otherwise use a file on the server, the entire file is sent from the server to that client. The three-tier approach would usually use the database server approach that involves splitting the process between the client and server and there is much less network data traffic in this approach.

The distributed database solves the problem with network shutdowns on a server and the cost for the shutdown. A centralized database concept is distributed which spreads the data among different cities on a distributed network, each of which has its own computer and data storage facilities.

With a distributed database plan, if one computer server goes down than a backup server in another location can take over to keep the data server up while the other is shutdown.

One of the main advantages for location a portion of a database in congested cities can be availability. If one site goes down, everyone else on the network can still access the site or backup while minimizing cost during data access.

XIII. CHAPTER 13 THE DATA WAREHOUSE

D The difference between Transaction Processing System (TPS) versus Decision Support Systems (DSS) is the functionality between both. TPS scenarios involve everyday application systems that support banking and insurance operations, managing the parts inventory on manufacturing assembly lines. The DSS process is more aimed at specific design to aid managers in decision-making tasks.

A data warehouse is a broad-based, shared database for management decision making that contains data that has been accumulated over time. Data warehouse systems can help collect data for support in various positions such as management or administration.

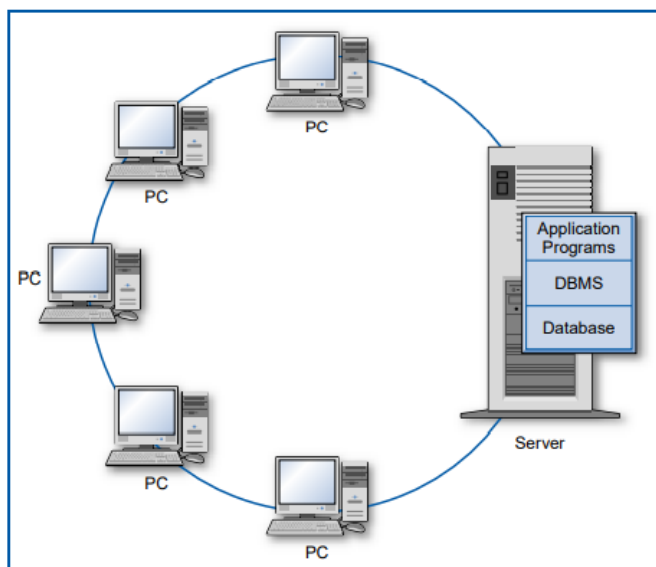
The Enterprise Data Warehouse (EDW) is a large-scale system that supports a variety of DSS applications and serve as a data resource. A full scale EDW is built around several different subjects since it incorporates data of an entire company. The Data Mart is more of a small-scale design for smaller organizations and usually companies would hold several data marts.

In some cases, data warehouses are often referred as multidimensional databases because each occurrence of the subject is referenced by an occurrence of each several dimensions or characteristics of the subject, one of which is time.

XIV. CHAPTER 14 DATABASES AND THE INTERNET

The Internet is associated with many factors for connecting to a database. Applications manage the data which can go through electronic commerce using a connected database. Data

is always expanding with new data types that will lead to new connectivity issues. Availability and scalability of databases are geared for change from the structure of databases and the amount of data presentable for use. Security and privacy analysis are applied to continue management of the data for daily usage/ Access to databases over the Internet has brought increased focus to operations. Data operations formatted with XML can help with data conversion for users to transact business over the Internet.



REFERENCES

1. John Wiley, "Fundamentals of Database Management", 2012
2. CHM Computer History Museum, "Memory & Storage", 2022
3. Bob Losey, "Solid State Drives (SSDs) vs Hard Disk Drives (HDDs) – What Is Best For My Next IBM i (iSeries/AS400) POWER Server?", 2019
4. Shanika Wickramasinghe, Muhammad Raza, "DBMS: Database Management Systems Explained", 2021